

# Nathan Ma

[nathanma.ca](http://nathanma.ca) | [linkedin.com/in/nathanma0](https://www.linkedin.com/in/nathanma0) | [n23ma@uwaterloo.ca](mailto:n23ma@uwaterloo.ca)

## EDUCATION

### University of Waterloo

BASc, Mechanical Engineering, Option in Artificial Intelligence

Sep 2024 - Present

GPA: 4.0

## EXPERIENCE

### Mechanical Systems Lead | Goose Robotics

Feb 2026 - Present

- Designing a **12-DOF quadruped robot** with a **PLA/aluminum hybrid frame**, using aluminum structural members for rigid load paths and passive thermal management of onboard electronics.
- Implemented **Watt 6-bar linkages** at each knee to keep servo mass proximal to the body, reducing per-leg rotational inertia and improving dynamic response during locomotion.
- Architected a hierarchical control system with a **Raspberry Pi** for high-level planning communicating over **UART** to a custom **STM32** servo driver generating **12-channel PWM** via HAL timers for low-level joint control.
- Developed per-leg geometric **inverse kinematics** from scratch using trigonometric methods, implementing open-loop **trot gait** with coordinated swing/stance trajectories across diagonal leg pairs.
- Integrating **RPLidar C1** with **ROS2** via the **rplidar\_ros** driver, implementing **SLAM Toolbox** for real-time 2D occupancy mapping and onboard environment perception.

### Mechanical Engineering Intern | Toyota Motor Manufacturing Canada

Jan 2026 - Apr 2026

- Led a TBP-driven **continuous improvement** project focused on inefficiencies with vision system enabled processes. Reduced shell-body sealer vision system fault downtime by **86%** through process optimization, system revalidation, and coordination with cross-functional production and maintenance teams.
- Designed a **poka-yoke locating guide** for door frame weld jigs via rapid prototyping in **SolidWorks**, constraining part seating geometry to eliminate misload-induced defects without impacting cycle time.
- Coordinated the commissioning, validation, and process optimization of new manufacturing equipment including vision systems, hoists, tooling, and **PLC**-controlled systems to improve cycle time, quality, cost, and safety.
- Calculated new **robotic weld schedules** and retaught robots to reduce weld spatter, minimize weld separation defects, and extend welding tip life.

### Industrial Engineering Intern | General Motors

May 2025 - Aug 2025

- Led a **continuous improvement** project focused on eliminating production inefficiencies through targeted **root cause analysis**, collaborating with **cross-functional teams** to optimize line balance. Successfully reduced job cycle times by up to **11.8%** and eliminated significant line downtime, directly resulting in increased shop output.
- Conducted **time studies** and redesigned standardized work across **30+** stations, applying **lean manufacturing** principles to identify wasted movement and ergonomic risks, reducing cycle time and ergonomic strain.
- Updated **10,000+ sqft** of factory layout using **AutoCAD**, streamlining cross-functional operations by delivering precise, standardized floor plans that enhanced coordination and efficiency across departments.
- Observed how **DFMA** decisions influence cost and ergonomics downstream on the plant floor—connecting design with real-world assembly outcomes.

## PROJECTS

### Internal Cycloidal Actuator | SolidWorks, 3D-Printing, Soldering

- Designed and built a fully integrated **cycloidal actuator** from scratch: custom hand-wound **36S42P outrunner BLDC** motor and internal cycloidal gearbox, modelled as a compact integrated unit in **SolidWorks**.
- Validated electromagnetic design by successfully spinning the motor under **ESC control**; currently integrating **Moteus** driver for closed-loop torque and position control with target **field-oriented control** over CAN bus.
- Diagnosed gearbox seizure from compounding tolerance stack-up due to PLA creep; concluded cycloidal drives exceed FDM precision capabilities and pivoted to a **planetary gearbox** planned top-down from first principles.

## TECHNICAL SKILLS

**Mechanical Design:** SolidWorks, CATIA 3DX, Siemens NX, Fusion 360, Autodesk Inventor, ANSYS, AutoCAD, GD&T, FEA, CFD, Tolerance Stack-Up, DFM, FMEA, Topology Optimization

**Fabrication & Prototyping:** Machining, 3D Printing (FDM, SLS, SLA), Laser Cutting, Welding, Soldering, Silicone Casting, Injection Molding, Rapid Prototyping

**Robotics & Embedded:** ROS2, STM32, Raspberry Pi, ESP32, Arduino, KiCad, CAN Bus, UART, PWM, Python, C++, MATLAB, Excel